GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Comprehensive GATE Notes: General Aptitude

Common Section: Verbal, Quantitative & Analytical Ability (15 Marks)

Subject Code: EC-GA • **Target Exam:** GATE ECE • **Weightage:** 15 Marks Fixed

1. Quantitative & Analytical Reasoning

- **Work and Rates:** 1/T_comb = 1/T_A + 1/T_B => T_comb = (T_A * T_B) / (T_A + T_B).
- **Clock Angle Formula:** Angle between hands at H hours and M minutes is theta = |30*H - 5.5*M| degrees.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Combined Work Time	**T_{comb} = (T_A * T_B) / (T_A + T_B)**	T_A, T_B = Individual Times	Harmonic rate addition for simultaneous tasks
Clock Hands Angle	**theta = |30 H - 5.5 M|**	H = Hours (1-12), M = Minutes (0-59)	Instantaneous acute angle between clock hands

GATE PYQ (2 Marks) - Pipe & Cistern Filling Time
Question: Pipe A fills tank in 12 hours, Pipe B empties in 18 hours. If both open, how long to fill completely?

Step 1: Net rate = 1/12 - 1/18 = (3 - 2)/36 = 1/36 tank/hr.
Step 2: Total time = **36 hours**.

Essential GATE Exam Pitfalls & Scoring Traps

- **Important:** General Aptitude carries 15% of the total GATE marks with very high scoring efficiency.